

Herpes Zoster: Clinical Features and Complications

Dermatomal Involvement and Rash Evolution

- **Early stage:** Characterized by erythema within a thoracic dermatome, followed by grouped vesicle formation.
- **Later stage:** Crusted lesions appear on the back (site of initial eruption), with newer, confluent hemorrhagic vesicles and bullae on the lateral chest wall. Vesicles extending beyond the dermatome suggest hematogenous dissemination, which is not uncommon.
- Crusting typically persists for 2–3 weeks.
- New lesions may appear for 1–4 days in immunocompetent individuals, occasionally up to 7 days.
- Rash severity and duration are greatest in older adults and mildest in children.

Ophthalmic Zoster

- Involves the ophthalmic division of the trigeminal nerve in 10%–15% of herpes zoster cases.
- Rash distribution extends from the eye to the skull vertex, with sharp midline termination on the forehead.
- Isolated involvement of the **supratrochlear** and **supraorbital** nerves usually spares the eye.
- **Nasociliary branch** involvement (which innervates the eye and nasal tip/side) increases risk of ocular complications.
- **Hutchinson's sign:** Lesions on the tip or side of the nose indicate potential ocular involvement, necessitating ophthalmologic evaluation.
- Ocular involvement occurs in 30%–40% of ophthalmic zoster cases.
- Common complication: **Neurotrophic keratitis** due to impaired corneal sensation, potentially leading to chronic ulceration.

Cranial Nerve Involvement

- Herpes zoster affecting the **mandibular (V3)** and **maxillary (V2)** divisions of the trigeminal nerve can cause oral, pharyngeal, or laryngeal lesions.
- **Ramsay Hunt syndrome:**
- Results from facial (VII) and auditory (VIII) nerve involvement.
- Clinical triad: facial palsy, vesicles on the external ear or tympanic membrane, and sensorineural symptoms (tinnitus, vertigo, hearing loss).
- May also present with intraoral lesions (e.g., on tongue, soft palate).

Pain in Herpes Zoster

- Pain is the primary symptom, especially in elderly patients.
- Acute phase (within 30 days of rash onset) is marked by dermatomal pain ranging from mild to severe.
- Descriptors include: burning, deep aching, tingling, itching, stabbing.
- Severe pain may be described as **excruciating** or **horrible**.
- Associated with:
- Reduced physical function
- Emotional distress
- Impaired social functioning

Herpes Zoster in Immunocompromised Hosts

- Immunocompromised individuals are at higher risk for severe complications beyond postherpetic neuralgia (PHN).
- **Cutaneous complications:**
- Skin necrosis and scarring
- Cutaneous dissemination (25%–50% incidence)
- **Visceral dissemination:**
- Occurs in ~10% of those with cutaneous spread
- Frequently involves lungs, liver, and brain

- Often fatal
- **HIV/AIDS patients:**
- Prone to multiple recurrences of herpes zoster as immune function declines
- Recurrences may occur in the same, different, or multiple contiguous/non-contiguous dermatomes
- Disease may be severe, with widespread cutaneous and visceral dissemination